arihantjaingoa@gmail.com
LinkedIn | GitHub

Arihant Jain

Education

University of California, San Diego — San Diego, CA
M.S. in Machine Learning and Data Science, Focus in Robotic Systems and Control — *Sep 2026 – Jun 2027*

University of California, San Diego — San Diego, CA
B.S. in Computer Engineering; GPA: 3.78/4.0 | Major GPA: 3.96/4.0 — *Sep 2022 – Jun 2026*

- **Coursework:** Machine Learning, Deep Learning, Computer Vision, Embedded Systems, Control Systems, Autonomous Sensing and Estimation, Planning and Optimal Control.

Work Experience

Existential Robotics Lab — San Diego, CA
Graduate Researcher — *Nov 2025 – Present*

- Developing PyTorch visual world models using 192D latent representations and temporal-distance quasimetric reachability for complex long-horizon manipulation and loco-manipulation tasks.
- Building latent-space RRT planners with trajectory interpolation to achieve above-state-of-the-art long-horizon multi-step goal success, low compute overhead, and real-time replanning under changing task goals without repeated model rollouts.
- Generating intermediate latent-space subgoals for visual goal-conditioned robotic tasks and using learned inverse dynamics to translate planned latent trajectories into executable actions for closed-loop goal tracking.
- Learning partially differentially flat approximations of complex robot dynamics to simplify trajectory optimization and planning in systems with nonlinear, high-dimensional motion constraints and coupled motion constraints.
- Developing sampling-based signed distance fields over learned latent dynamics to efficiently approximate feasibility and support motion planning in high-degree-of-freedom configuration spaces in sampling-based search.

Cadel.AI — Bengaluru, Karnataka
Software Developer Intern — *Jul 2026 – Sep 2026*

- Raised success rates from 69% to 99.1% across 15+ financial workflows by designing deterministic tools for document parsing and execution, reducing reliance on error-prone language-model steps.
- Benchmarked Gemini, Claude, Kimi, Codex, and open-source models to identify workflow failures and token waste.
- Applied GRPO-based reinforcement learning to improve workflow execution consistency by 1.7× across complex financial tasks and deployed the resulting improvements to production.

Jacobs School of Engineering / Rady Children's Hospital — San Diego, CA
Research Intern — *Jan 2025 – Jun 2026*

- Redesigned a multimodal clinical glove containing 725 force-sensitive resistive sensors and a MotionNode IMU to capture pressure and inertial signals for quantitative patient movement analysis.
- Built a 100 Hz multithreaded Python acquisition pipeline, replacing MATLAB for glove sensor collection.
- Evaluated HDBSCAN on movement data from 50 patients; presented findings at UCSD SRIP 2025.

Projects

Triton Robotics — San Diego, CA
Embedded Systems Lead — *Jan 2025 – Jun 2026*

- Led a 10–15 person embedded systems team within an 80-member robotics organization, mentoring new recruits, coordinating firmware development, and reviewing C++ code for reliable embedded actuation and integration across robot subsystems.
- Used MATLAB system identification and PID autotuning to reduce 170° yaw settling time from 1.2 s to <400 ms, improving closed-loop heading response for autonomous robot motion and faster stabilization after commanded large-angle rotations.
- Developed real-time C++ firmware on STM32F446 microcontrollers and integrated BNO055 IMUs and peripheral interfaces over I2C, SPI, CAN, and UART to support real-time sensing, actuator communication, and reliable low-level robot control.
- Implemented Point-LIO LiDAR-inertial odometry and localization using Odin 1 and Unitree L2 LiDARs in a ROS2 Humble autonomy stack for state estimation and navigation for mobile robot pose tracking.
- Built EKF-based pose prediction and combined vector-field navigation with computer-vision obstacle avoidance for autonomous motion planning and closed-loop robot navigation in obstacle-rich operating environments.

CAD2Sim – CAD-to-Simulation Compiler | GitHub — Oct 2026

- Built a deterministic Python compiler that converts Onshape, SolidWorks, Fusion, Creo, and STEP robot assemblies into URDF/SRDF, MJCF, and USD assets for MuJoCo, Isaac Lab, Gazebo, and other simulators.
- Implemented geometric joint inference, collision mesh decomposition, sampled self-collision matrices, and configuration-driven robot model generation to turn heterogeneous CAD exports into simulation-ready kinematics and geometry.
- Built simulation validation and cross-simulator behavior checks alongside a CAD2Sim agent workflow that iteratively inspects, tests, and refines generated robot assets before use in robotics planning, control, and simulation experiments.

AGNTCity – SanD Hacks Multi-Agent Systems Track Winner | Project — Feb 2026

- Co-developed AGNTCity, winner of the SanD Hacks Multi-Agent Systems track, in under 24 hours; the system converts trending Moltbook agent discussions into human-readable news articles through scheduled supervisor-worker pipelines.
- Integrated an AGNTCY-based multi-agent architecture with MCP tools for contextual retrieval and caching, agent-to-agent communication over NATS, LangGraph/LangChain orchestration, FastAPI services, and a React frontend.

Skills

Languages: Python, C++, C, CUDA, MATLAB, Simulink, JavaScript/TypeScript
ML / Robotics: PyTorch, JAX, World Models, LeWM, JEPA, Reinforcement Learning, Computer Vision, ROS2, MoveIt, OMPL, RRT, MuJoCo
Controls / Embedded: PID Control, System Identification, STM32, RTOS, IMU, I2C, SPI, CAN, UART, LiDAR, Sensor Fusion, EKF, Point-LIO
Tools: Linux, Git, Docker, OpenCV, HDBSCAN, SDFs, FastAPI, LangGraph, MCP, NATS